

Credit-Wise Loan Approval System

SecureTrust Bank processes hundreds of daily personal and home loan applications across India. The legacy manual review process faced significant operational challenges: it was slow, subjective, and prone to error. This led to two major business risks:

Financial Loss (High-Risk Approvals): High-risk applicants mistakenly receiving approval, driving up default rates.

Revenue Loss (Creditworthy Rejections): Low-risk, creditworthy applicants being wrongfully rejected, leading to customer churn and missed revenue.

To solve this, an automated, machine-learning-driven decision pipeline was developed to evaluate applicant records and output an automated recommendation (Approved vs. Rejected) prior to final administrative sign-off.

Business Optimization Hierarchy

- **Primary Objective:** Minimize False Positives (approving high-risk borrowers) to mitigate direct financial defaults.
- **Secondary Objective:** Minimize False Negatives (rejecting creditworthy applicants) to maximize loan origination revenue.

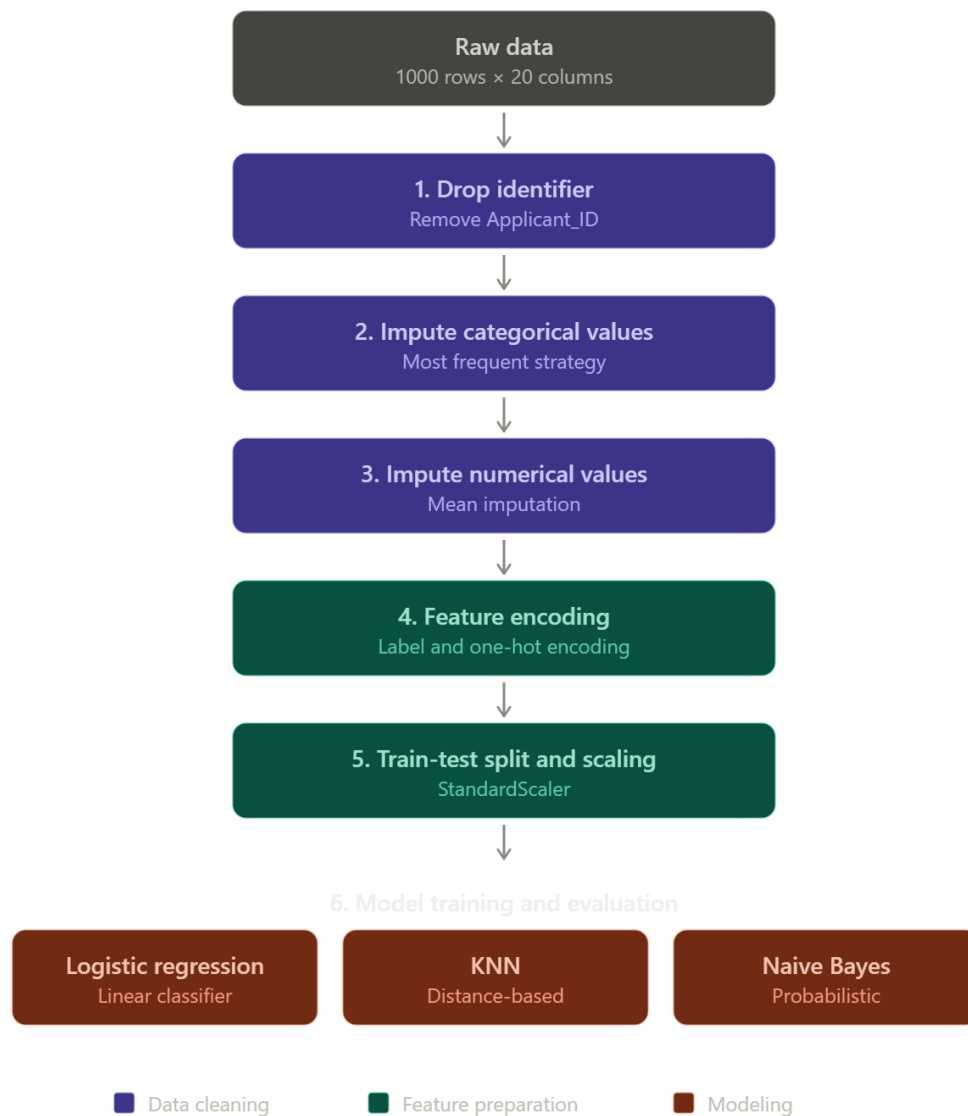
Dataset Description & Schema

The dataset comprises **1,000 historical applicant records** across **20 initial features** covering personal, employment, debt, collateral, and bureau data.

No.	Feature	Description
1	Applicant	ID of the applicant
2	Applicant_Income	Income of the applicant per month
3	Coapplicant_Income	Income of the co-applicant per month
4	Employment_Status	3 options (salaried/ self-employed/ business)
5	Age	Applicant age
6	Marital_Status	2 options (married/ single)
7	Dependants	Number of dependents on applicant income
8	Credit_Score	Credit-bureau-score

9	Existing_Loans	Number of already running loans
10	DIT_Ratio	Debt-to-income ratio (this debt is the make sure you have enough money to pay for your debt and how much is left with you, the bank uses DTI to calculate if that leftover amount is big enough to safely fit a new loan payment)
11	Savings	Your saving balance
12	Collateral_Value	Collateral value is the estimated cash worth of an asset that you pledge to a bank to secure a loan.
13	Loan_Amount	Loan amount requested
14	Loan_Term	Loan duration (months)
15	Loan_Purpose	4 categories (home/education/personal/business)
16	Property_Area	3 options (Urban/Semi-Urban/Rural)
17	Education_level	3 options (Graduate/Postgraduate/Undergraduate)
18	Gender	Male/Female
19	Employer_Category	3 options (Govt/Private/Self)
20	Loan_Approved(Target)	yes=Approved,no=Rejected

Step-by-Step Data Preprocessing Pipeline



1. Feature Drop & Noise Reduction

- Applicant_ID was removed as it serves purely as an index and carries no predictive signal for creditworthiness.

2. Missing Value Treatment

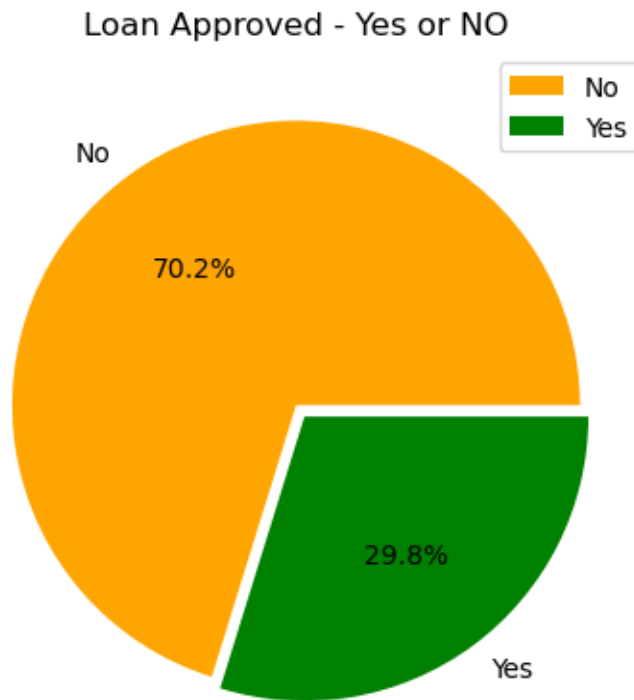
Data profiling revealed exactly **50 missing values per feature column** across all 1000 records. An automated imputation strategy was applied using `sklearn.impute.SimpleImputer`:

- **Categorical Imputation:** Categorical features (Employment_Status, Marital_Status, Loan_Purpose, Property_Area, Education_Level, Gender, Employer_Category, Loan_Approved) were imputed using the **mode** (strategy='most_frequent').
- **Numerical Imputation:** Continuous attributes (Applicant_Income, Coapplicant_Income, Age, Dependents, Credit_Score, Existing_Loans, DTI_Ratio, Savings, Collateral_Value, Loan_Amount, Loan_Term) were imputed using column **means** (strategy='mean').

3. Class Imbalance Identification

Inspection of the target feature `Loan_Approved` revealed an imbalanced distribution:

- **Rejected (No):** 702 instances (~70.2%)
- **Approved (Yes):** 298 instances (~29.8%)

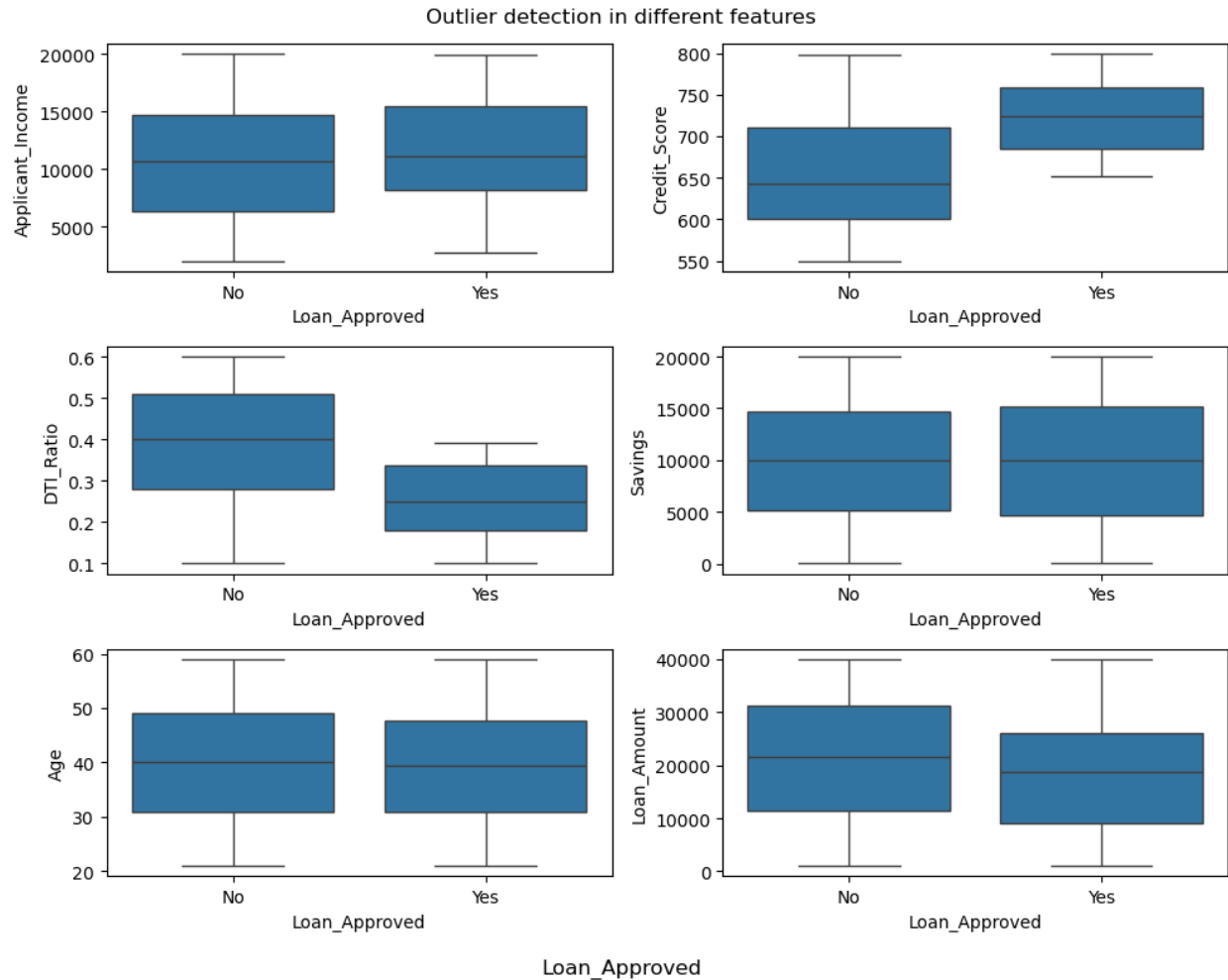


Exploratory Data Analysis (EDA) & Visualization Integration

The Exploratory Data Analysis focused on identifying structural patterns, evaluating feature relationships with the target variable (Loan_Aproved), and detecting anomalies or statistical noise prior to model execution.

1. Outlier Detection Across Key Numerical Features

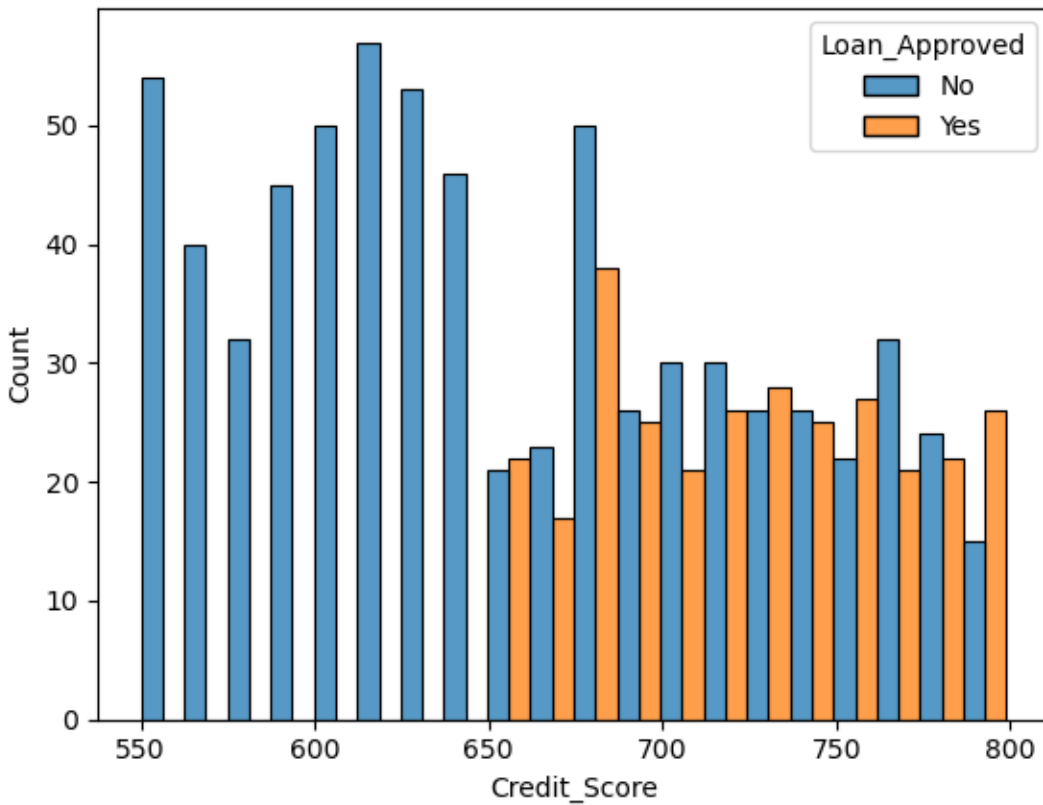
A multi-panel boxplot analysis was conducted on the six primary continuous drivers: Applicant_Income, Credit_Score, DTI_Ratio, Savings, Age, and Loan_Amount, grouped against Loan_Aproved.



Key Takeaway: No significant or destructive outliers were detected across any of the key numerical variables.

2. Credit Score Distribution & Approval Sensitivity

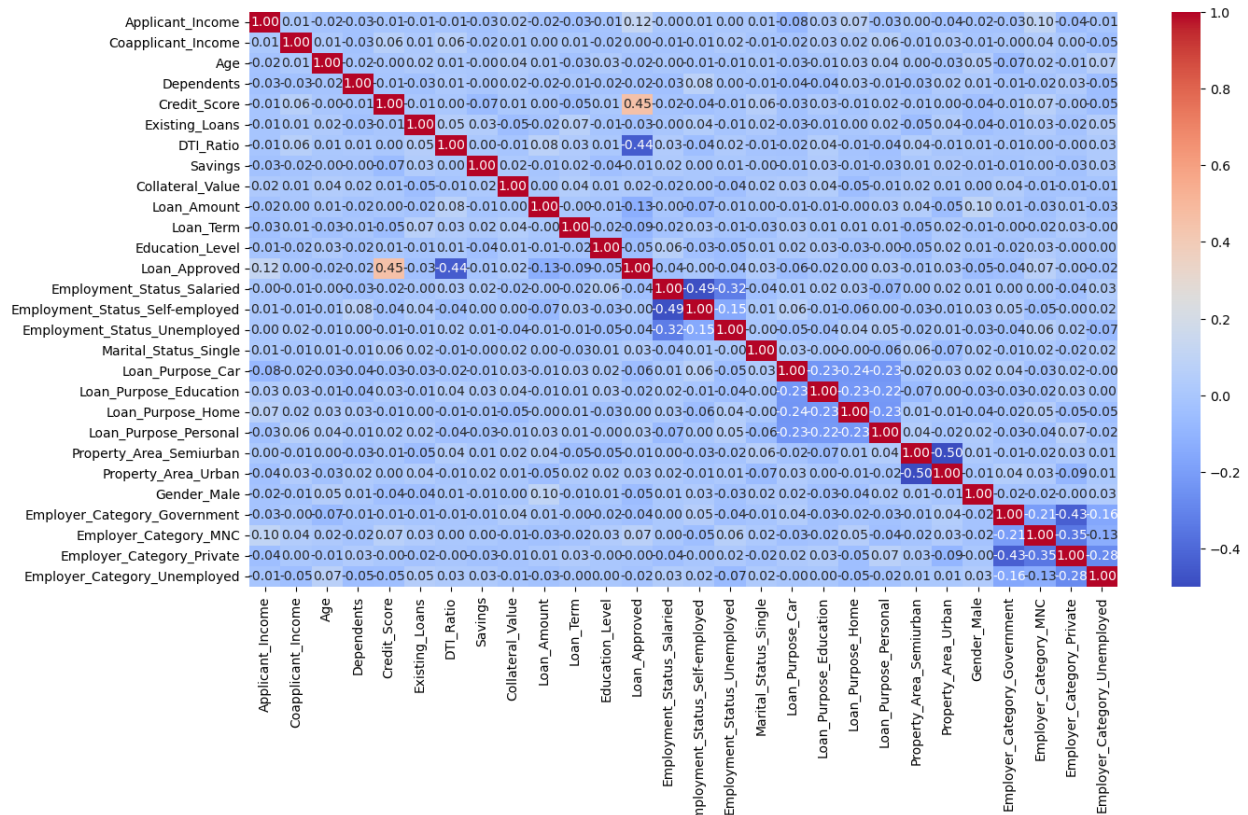
A dodge-style histogram (bins=20) was generated to inspect the exact relationship between an applicant's Credit_Score and their approval outcome.



Key Takeaway: Credit_Score exhibits a strong direct relationship with Loan_Approved.

3. Correlation Matrix & Interdependency Analysis

A complete numerical correlation matrix was computed and rendered as an annotated heatmap to map linear dependencies against the target variable Loan_Approved.



Primary Correlations with Loan_Approved:

- Positive Correlation — Credit_Score:** Shows the strongest positive correlation with Loan_Approved, reinforcing that higher bureau scores directly increase loan qualification probability.
- Negative Correlation — DTI_Ratio:** Displays a strong negative correlation with Loan_Approved, demonstrating that higher existing debt obligations relative to income significantly increase the likelihood of loan rejection.

Feature Transformation & Model Development Roadmap

To finalize the predictive engine, the project execution follows these remaining implementation steps:

1. Categorical Encoding & Feature Scaling

- **Target Encoding:** Binary conversion of Loan_Approved (Yes $\rightarrow$ 1, No $\rightarrow$ 0).
- **Feature Encoding:** One-Hot Encoding (OneHotEncoder) for nominal attributes (Property_Area, Loan_Purpose, Employer_Category) and Label Encoding (LabelEncoder) for binary/ordinal attributes.
- **Feature Scaling:** Application of StandardScaler to transform continuous numeric variables, ensuring zero mean and unit variance. This prevents distance-sensitive algorithms like KNN and gradient-based solvers like Logistic Regression from being distorted by feature magnitudes (e.g., Applicant_Income vs. DTI_Ratio).

Model Training, Hyperparameter Tuning & Evaluation

The standard preprocessed and scaled dataset ($N = 1,000$) was partitioned using an 80/20 stratified train-test split, yielding a holdout evaluation set of **200 test samples** (139 actual rejections, 61 actual approvals). Three core classification algorithms were trained and evaluated: **Logistic Regression**, **K-Nearest Neighbors (KNN)** (both default and hyperparameter-tuned), and **Gaussian Naive Bayes (GaussianNB)**.

Model Architecture	Precision	Recall	Accuracy	False Positives (FP)	False Negatives (FN)
Logistic Regression	78.33%	77.05%	86.50%	13	14
KNN (Default)	62.75%	52.46%	76.00%	19	29
KNN (Tuned: $k=13$)	73.17%	49.18%	79.00%	11	31
Gaussian Naive Bayes	80.36%	73.77%	86.50%	11	16

Final Model Selection & Business Recommendation

Gaussian Naive Bayes is selected as the production model for the SecureTrust Bank CreditWise Loan Approval System.

1. **Minimizing Direct Financial Loss (Primary Goal):** Naive Bayes achieved the highest Precision (**80.36%**), tying with tuned KNN for the lowest False Positive rate ($\text{FP} = 11\%$). This minimizes default exposure by ensuring high-risk borrowers rarely bypass automated rejection.
2. **Preserving Loan Origination Revenue (Secondary Goal):** Unlike tuned KNN (which dropped Recall to 49.18%), Naive Bayes maintains a strong Recall of **73.77%** ($\text{FN} = 16\%$). This avoids customer churn by ensuring creditworthy borrowers are reliably approved.
3. **Balanced Efficiency:** Delivering a top-tier Accuracy of **86.50%** alongside the lowest combined error profile ($\text{FP} + \text{FN} = 27\%$), Naive Bayes offers the most effective decision threshold for operational deployment.

Model Deployment & Interactive Web Application (Streamlit)

To transition the trained Gaussian Naive Bayes model into an operational business tool, a lightweight web application was developed using **Streamlit** (`app.py`). This interactive interface allows you to input applicant parameters and receive instant, automated risk assessments and approval recommendations.



CreditWise Loan Approval System

Fill in the applicant's financial and personal details below to predict loan approval status.

Applicant Details

Applicant Income (₹)

100000.00

- +

Loan Amount (₹)

500000.00

- +

Coapplicant Income (₹)

200000.00

- +

Loan Term (Months)

36

- +

Age

25

- +

Existing Loans

0

- +

Dependents

0

- +

Savings (₹)

555000.00

- +

Education Level

Undergraduate



Loan Purpose

Education



Employment Status

Salaried



Property Area

Semiurban



Marital Status

Married



Employer Category

Private



Gender

Male



Evaluate Application



Loan Application Status: APPROVED